

EasyNet 状态机器完整设计

Join、Resolve、Invoke、Read Model 与 Failure 可追踪契约

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Abstract

本文定义 EasyNet 从前端 pairing token 到 easynet join、CLI daemon 接入 Hub/Axon runtime、Backend/Frontend 显示 connected 的完整状态机器, 同时覆盖 RFC-005 URA namespace resolution 迁移后的 ability resolve、invoke、read model 和 failure propagation。核心目标是: 每个用户可见状态都有稳定状态码; 每个失败都能定位到具体 transition; 所有 invocation 必须先经过 Axon ResolveAnswer.FinalRoute, 再按 resolver-selected next hop dispatch。

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1 结论

Invariant

[axiom] credential accepted != daemon connected != namespace visible != frontend connected.

Invariant

[axiom] bare ability name 只能作为 resolver query input。Backend prepare、Frontend signing、CLI daemon dispatch 和 signed Invocation 必须使用 ResolveAnswer.FinalRoute 中的 canonical ability_ura、route_ura、dispatch_name 和 next_hop。

当前问题不是单一 HTTP 页面、单一 ability 或单一 handler 的 bug, 而是 RFC-005 迁移尚未完全收敛后的系统性状态机器缺口。症状包括: join 保存 credential 后 daemon 连接失败却曾被当作成功; terminal.list 因 device 不在 PresenceRegistry 返回 fallback; pages.list 因 agent 未 advertised 变成普通 500; history/list 因缺 canonical route 返回裸 400; meta.list_resources 缺 resolver catalog route。

2 仓库边界

层	应拥有	不应拥有
EasyNet-Axon	URA 语法、ResolveQuery、ResolveAnswer、route evidence、negative taxonomy、failure code、InvocationReceipt.failure	Backend 产品 DB、Frontend UX、CLI 本地插件实现
EasyNet-Cli daemon	local device abilities、hosted agent runtime、Axon bridge/session、device namespace authority、resolve-before-invoke、local receipt producer	Backend HTTP policy、Frontend connected 判定
EasyNet Backend	pairing lifecycle、credential issuance、user ownership、read model projection、typed unavailable DTO、signed invoke prepare/submit facade	URA canonicalization fork、owner/route inference、Axon resolver negative 字符串解析
EasyNet Frontend	token UX、SSE cache invalidation、typed state rendering、actionable error display	liveness truth、ability route builder、resolver fallback

Decision

[derived] Backend 经 daemon.sock 调 Hub/daemon ability 是合法 transport-binding; Backend 在 Go 里 fork Axon 协议契约是不合格实现。Frontend 只能消费 API routes/read model, 不能拼 canonical ability URA。

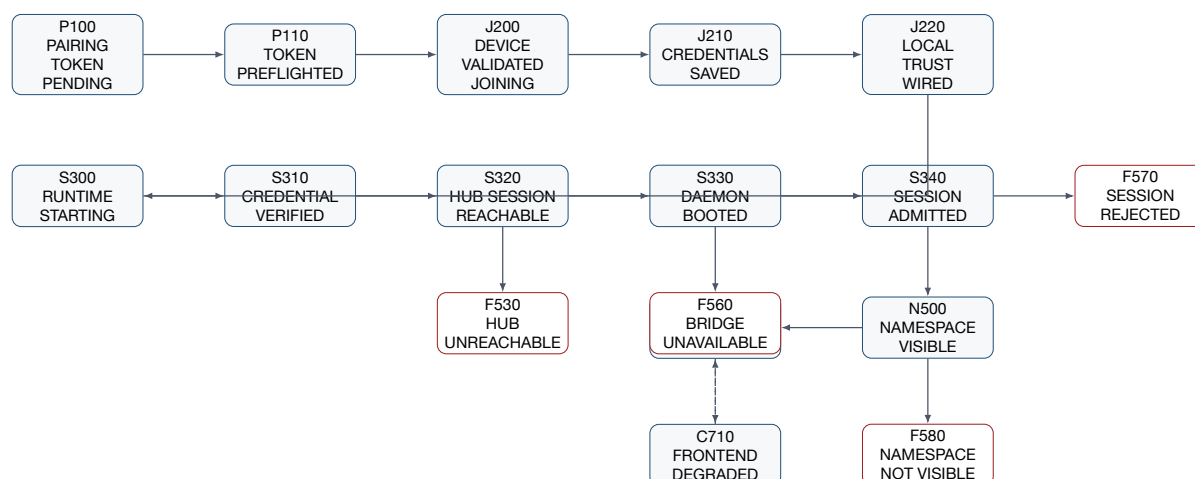
3 状态码规则

前缀	语义
P	Pairing token lifecycle
J	Join credential lifecycle
S	Session/runtime lifecycle
N	Namespace visibility/readiness
C	Frontend connected/degraded UI
R	Resolver route selection
I	Invocation prepare/submit/execution
M	Mission/EAL composite invocation orchestration
D	Download/body-committed stream lifecycle
O	Operator runtime stop lifecycle
U	Self-uninstall/leave lifecycle
A	Ability catalog/read model
F	Failure terminal or degraded state

状态码必须稳定、可定位、可追踪、可组合。稳定表示 UI、CLI、日志、测试不得依赖人类字符串。可定位表示 failure 必须携带 transition、from_state、interrupted_state。可组合表示列表 API 可以返回 product rows, 同时携带 resolve_unavailable[]。

4 Join 到 Connected 状态机器

4.1 状态图



4.2 状态表

Code	State	Owner	Meaning	Terminal
P100	PAIRING_TOKEN_PENDIN G	Backend	token 已创建, hash 已保存, TTL 生效	No
P110	PAIRING_TOKEN_PREFLI GHTED	Backend / CLI	CLI 证明持有 token, 取得 reserved node/trust material	No
F120	PAIRING_TOKEN_EXPIRE D	Backend	token 过期	Yes

F130	PAIRING_TOKEN_CONSUMED	Backend	token 已被使用或状态非 pending	Yes
J200	DEVICE_VALIDATED_JOINING	Backend	credential issued, DB 状态 validated, liveness 未证明	No
J210	CREDENTIALS_SAVED	CLI	credential/config/key material 已落本地	No
J220	LOCAL_TRUST_WIRED	CLI	realm trust 和 federation peer 写入完成	No
S300	RUNTIME_STARTING	CLI	正在启动 daemon	No
S310	HUB_CREDENTIAL_VERIFIED	CLI / Backend	Backend 接受 credential token	No
S320	HUB_SESSION_ENDPOINT_REACHABLE	CLI	Hub session endpoint TCP/TLS 可达	No
S330	DAEMON_BOOTED	CLI	daemon.sock up, runtime process alive	No
S340	SESSION_ADMITTED	CLI daemon / Hub	<self>.session admission 成功	No
N500	NAMESPACE_VISIBLE	Axon / Hub / CLI	device URA 已进入 resolver / PresenceRegistry 可见面	No
C700	FRONTEND_CONNECTED	Backend / Frontend	后端 read model 投影 ONLINE, 前端显示 connected	No
C710	FRONTEND_DEGRADED	Backend / Frontend	DB row 可显示, runtime resolver 有 typed unavailable	No
C790	FRONTEND_OFFLINE	Backend / Frontend	session/heartbeat lost, 或 PresenceRegistry 不含该 device	No

4.3 Failure 表

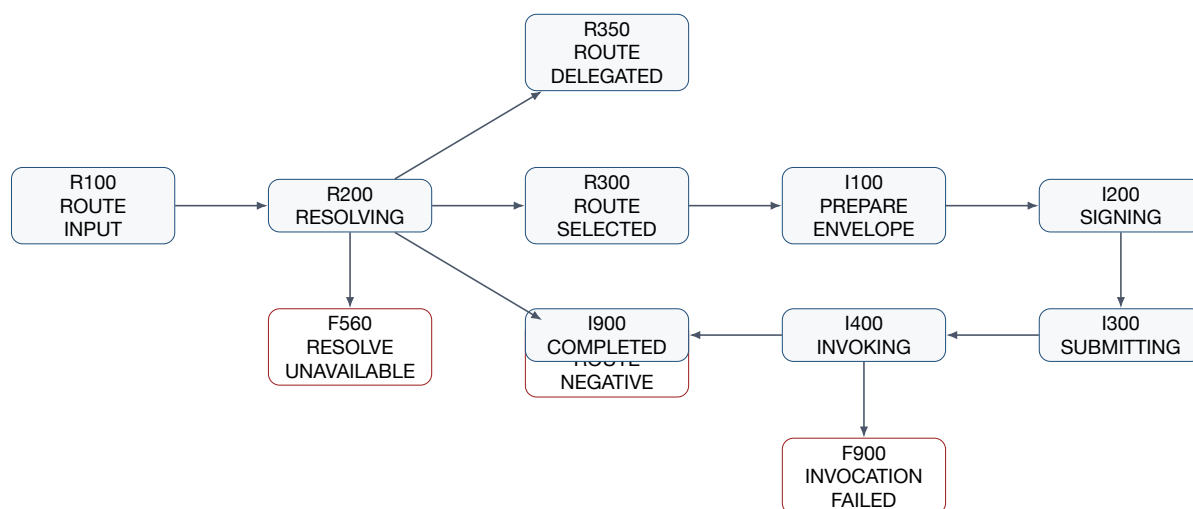
Code	Failure	Interrupted transition	Source	Operator action
F530	HUB_UNREACHABLE	S310 -> S320	CLI socket preflight	检查 Hub TLS endpoint, port, trust, compose
F541	CREDENTIAL_REJECTED	S300 -> S310	Backend credential verify	重新 join 或刷新 credential
F550	DAEMON_BOOT_FAILED	S300 -> S330	CLI daemon supervisor	看 daemon log
F560	BRIDGE_UNAVAILABLE	S330 -> S340	CLI Axon bridge loader	设置 EASYNET_DENDRITE_BRIDGE_LIB/HOME
F570	SESSION_ADMISSION_REJECTED	S340 admission	Hub admission / Axon	检查 caller signature, trusted key, realm trust
F580	NAMESPACE_NOT_VISIBLE	S340 -> N500	resolver / PresenceRegistry	检查 session 是否真的 admitted

Failure Contract

用户给出的 bridge missing 错误应记录为 F560 BRIDGE_UNAVAILABLE, interrupted transition 是 S330_DAEMON_BOOTED -> S340_SESSION_ADMITTED, 而不是 join 成功。

5 Resolve / Invoke 状态机器

5.1 状态图



5.2 Resolve 状态表

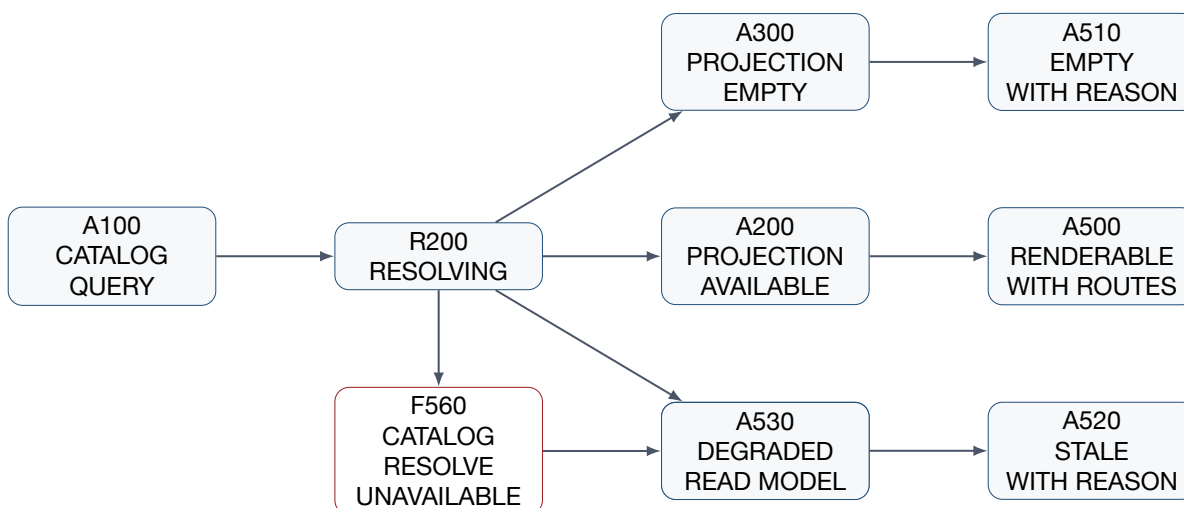
Code	State	Owner	Required data
R100	ROUTE_INPUT	Backend / Frontend / CLI caller	target_ura, ability_name, caller, subject, scope
R200	RESOLVING	CLI daemon / Axon	normalized ResolveQuery
R210	RESOLVING_PEER	Hub / Axon peer	delegation proof, authoritative realm
R300	ROUTE_SELECTED	Axon	ResolveAnswer.FinalRoute with selected next hop
R350	ROUTE_DELEGATED	Axon	peer hub delegation, no local dispatch
R400	ROUTE_NEGATIVE	Axon	NegativeReason, code, retry policy
F560	RESOLVE_UNAVAILABLE	CLI / Backend	transport/runtime failure locator

5.3 Invocation 状态表

Code	State	Owner	Invariant
I100	PREPARING_ENVELOPE	Backend	只能使用 R300.selected_route.ability_ura
I200	SIGNING	Frontend / client key	签名输入必须是 backend prepared canonical envelope
I300	SUBMITTING	Backend	submit body 必须携带 selected route/evidence
I400	INVOKING	CLI daemon / Axon runtime	dispatch only with resolver-selected next hop
I900	COMPLETED	Runtime producer	InvocationReceipt.state=Completed, 不带 failure

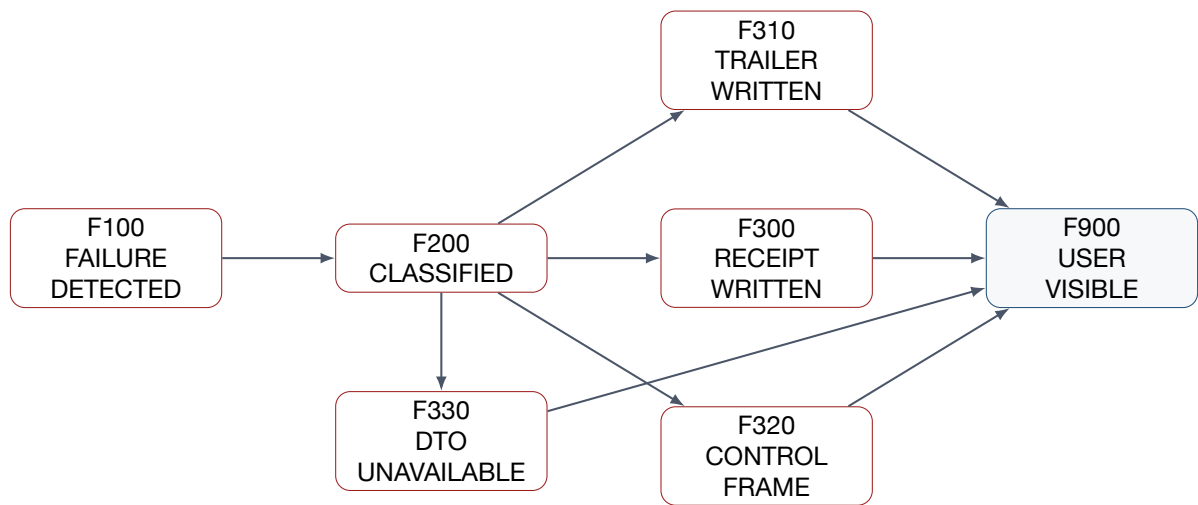
F900	INVOCATION_FAILED	Runtime producer	InvocationReceipt.failure 必须 populated
F901	INVOCATION_TIMED_OUT	Runtime producer	timeout reason 写入 typed failure
F902	INVOCATION_CANCELLED	Runtime producer	cancellation reason 写入 typed failure

6 Ability Catalog / Read Model 状态机器



Rule	Requirement
Catalog authority	device-local namespace authority; Hub stores signed projection/read model
Backend source	typed ResolveAnswer.records, not host placement inference
Frontend source	API routes[] and resolve_unavailable[], not local route construction
Empty list	only healthy when resolver returns positive empty or typed NODATA
Failure	resolve_unavailable[] must be populated for runtime/transport/negative failure

7 Failure Propagation 状态机器



Listing 1: FailureLocator

```
{
  "code": "TARGET_NOT_IN_PRESENCE_REGISTRY",
  "state_code": "F570",
  "source": "daemon_grpc.invoke_remote",
  "layer": "admission",
  "transition": "I300_SUBMITTING -> I400_INVOKING",
  "from_state": "I300_SUBMITTING",
  "interrupted_state": "I400_INVOKING",
  "message": "target device is not in PresenceRegistry",
  "retryable": true,
  "evidence": {
    "target_ura": "easynet:///r/localhost/device/<id>",
    "ability_ura": "easynet:///r/localhost/ability/device.<id>.terminal.list"
  }
}
```

Surface	Failure carrier	Rule
Unary HTTP before response	JSON	status code may be 4xx/5xx,
HTTP read model	{error,failure}	product reason must be typed
	resolve_unavailable	never return healthy empty list when resolver failed
HTTP download after headers	X-EasyNet-Failure trailer	base64url JSON FailureLocator, body remains octet-stream
WebSocket terminal	terminal_error	frontend renders typed
	control frame	code/source/message
Invocation receipt	InvocationReceipt.failed	only Failed/TimedOut/Cancelled receipt writes typed failure

8 核心数据结构

8.1 ResolveQuery


```
{
  "target_ura": "easynet:///r/localhost/device/<device-id>",
  "ability_name": "terminal.list",
  "subject_ura": "easynet:///r/localhost/user/<user-id>",
  "caller_ura": "easynet:///r/localhost/user/<user-id>",
  "scope": "TargetOwner",
  "include_abilities": false,
  "release_profile_min": "AuthoritativeLocal"
}
```

8.2 ResolveAnswer.FinalRoute

```
{
  "answer_kind": "FinalRoute",
  "selected_route": {
    "callee_ura": "easynet:///r/localhost/device/<device-id>",
    "ability_ura": "easynet:///r/localhost/ability/device.<device-id>.\n    terminal.list",
    "route_ura": "easynet:///r/localhost/route/<route-id>",
    "dispatch_name": "terminal.list",
    "next_hop": {
      "kind": "LocalDeviceAbility",
      "target_ura": "easynet:///r/localhost/device/<device-id>"
    }
  },
  "route_evidence": {
    "authority": "namespace.resolve",
    "selection_algorithm": "deterministic-weighted",
    "selection_seed_hash": "sha256:...",
    "profile": "Production"
  }
}
```

8.3 Backend Read Model DTO

```
{
  "items": [],
  "routes": [],
  "resolve_unavailable": [
    {
      "source": "namespace.resolve",
      "reason": "NOROUTE",
      "query_name": "terminal.list",
      "message": "target device is not in PresenceRegistry",
      "code": "TARGET_NOT_IN_PRESENCE_REGISTRY",
      "stage": "resolve",
      "retryable": true
    }
  ]
}
```

9 Doctor / Status 输出契约

Field	Required meaning
state_code	stable product state code
state	human-readable state
phase	pairing/join/runtime/namespace/frontend/invoke
transition	last completed or interrupted transition
failure.code	canonical failure code if failed/degraded
failure.source	source component
failure.retryable	operator can retry or must rejoin
next_action	concrete command or remediation

Listing 2: doctor human output target

EasyNet doctor

- device pairing paired as c130295e-9682-499e-bc09-72d4678a5887
- local runtime failed
 - state_code: F560
 - state: BRIDGE_UNAVAILABLE
 - transition: S330_DAEMON_BOOTED -> S340_SESSION_ADMITTED
 - reason: bridge library not found
 - next: set EASYNET_DENDRITE_BRIDGE_LIB or EASYNET_DENDRITE_BRIDGE_HOME, then run easynet runtime start
- federation peers configured, namespace not visible

10 HTTP / Frontend 状态渲染契约

Case	HTTP status	Body rule
resolver typed negative for list/read model	200	rows plus resolve_unavailable[], unless request itself invalid
missing canonical ability	400	typed failure with R400 or F560, not bare ability_ura is required
route in invocation prepare target offline during invoke	409/503	typed failure, code TARGET_NOT_IN_PRESENCE_REGISTRY
unauthorized policy	403	typed negative UNAUTHORIZED
healthy empty list	200	empty rows and no resolve_unavailable[]

11 Invocation 全量变体矩阵

11.1 维度

invoke 不是单一路径, 而是 resolver-selected route 加 transport profile 的组合。完整状态机必须覆盖 locality、tenant boundary 和 transport profile 三个维度。

Dimension	Values	Contract
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Locality	same device, same realm remote device, cross realm peer, hosted agent, hub/system ability	caller 不能从 locality 推断 route, 必须看 <code>ResolveAnswer.FinalRoute.next_hop</code>
Tenant boundary	same user, cross user same realm, cross user cross realm, service/system subject	policy result 必须进入 resolve answer 或 admission failure, 不能 在 frontend 静默过滤
Transport profile	unary, streaming, bidi, terminal/websocket, download-after-commit, Mission/EAL composite	transport 只能改变 failure carrier, 不能改变 canonical failure shape

11.2 Route variants

Variant	Predicate	Resolver answer	Dispatch next hop	Canonical failures
V0_LOCAL_SAME_DEVICE	caller、subject、 callee 在同一 device runtime	<code>FinalRoute(LocalDeviceAbility)</code>	local ability registry	NXDOMAIN, NODATA, UNAUTHORIZED
V1_SAME_REALM_SAME_USER_REMOTE_DEVICE	同 realm、同 user、 目标 device 在线	<code>FinalRoute(RemoteDeviceAbility)</code>	hub <code><self>.invoke_remote</code>	TARGET_NOT_IN_PRESENCE_REGISTRY, NOROUTE, STALE
V2_SAME_REALM_CROSS_USER	同 realm、跨 user 共享资源或授权 ability	<code>FinalRoute</code> or <code>Negative(UNAUTHORIZED)</code>	hub remote route	UNAUTHORIZED, SUBJECT_KEY_UNREGISTERED, DELEGATION_INVALID
V3_CROSS_REALM	target realm 与 caller realm 不同	Delegation then peer <code>FinalRoute</code>	trusted peer hub	REALM_NOT_TRUSTED, PEER_HUB_UNAVAILABLE, DELEGATION_REQUIRED
V4_HOSTED_AGENT_LOCAL_DEVICE	ability 属于本机 hosted agent	<code>FinalRoute(HostedAgentAbility)</code>	local agent runtime	AGENT_NOT_ADVERTISED, AGENT_NOT_RUNNING
V5_HOSTED_AGENT_REMOTE_DEVICE	hosted agent 在远端 device	<code>FinalRoute(RemoteHostedAgentAbility)</code>	hub to owning daemon	TARGET_NOT_IN_PRESENCE_REGISTRY, AGENT_NOT_ADVERTISED, OVERLOADED
V6_HUB_OR_SYSTEM_ABILITY	hub/system-owned ability	<code>FinalRoute(HubAbility)</code>	hub local ability	UNAUTHORIZED, ABILITY_QUERY_CONFLICT, THROTTLED
V7_BACKEND_PUBLIC_RESOURCE	pages/files/openai 等产品资源	<code>FinalRoute(ProductResourceRoute)</code>	route-selected host	ROUTE_PROFILE_NOT_AUTHORITATIVE, AGENT_NOT_ADVERTISED, NOROUTE

11.3 Transport 子状态机器

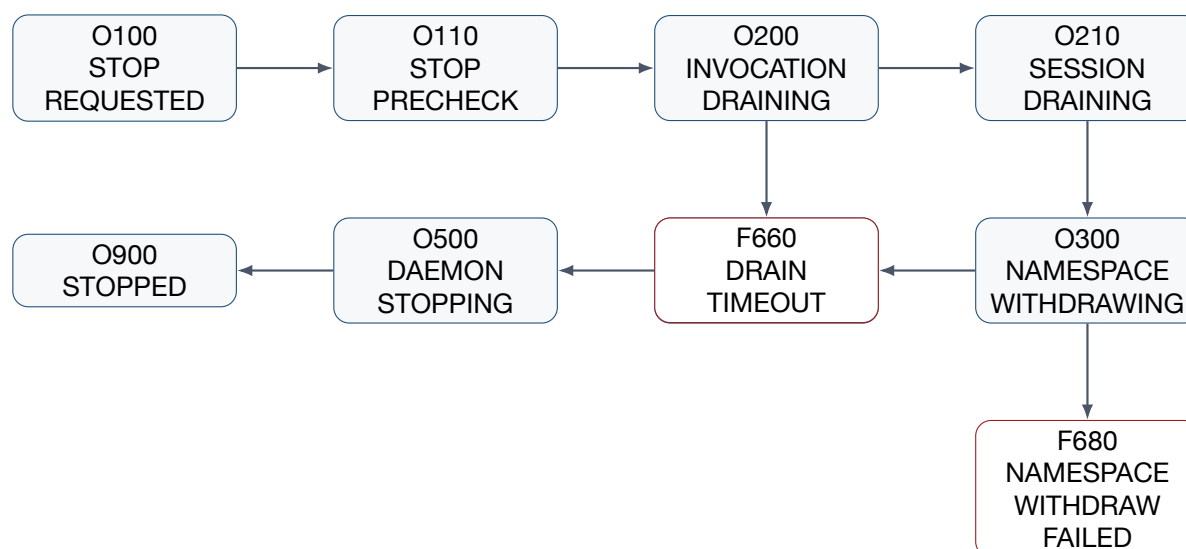
Profile	States	Failure carrier	Rule
Unary	I100 -> I200 -> I300 -> I410 -> I900/F900	HTTP JSON or receipt	response 未提交时使用 typed body
Server stream	I420 -> I421 -> I422 -> I900/F901	receipt/control frame	EOF 不能替代 typed close reason
Bidi stream	I430 -> I431 -> I432 -> I900/F902	receipt plus control frame	open failure 和 midstream failure 分开编码
Terminal/WebSocket	I420/I430 -> I421/I431 -> F320/I900	terminal_error frame	frontend 显示 code/source/message
Download after commit	D100 -> D200 -> D300 -> I900/D910	X-EasyNet-Failurebody 已提交后由 trailer	trailer 携带 failure
Mission/EAL composite	M100 -> M200 -> M300 -> M900/F910	parent receipt	child failure 保留, 不 flatten

11.4 Policy matrix

Caller/subject case	Same realm	Cross realm	Required evidence
Same user, same device	local positive route	delegation not needed	device credential, session id
Same user, remote device	remote positive route if online	peer positive route if trusted	ownership ledger, target presence
Cross user, explicit share	positive route only with grant	peer route only with cross-realm grant	grant id, subject URA, caller signature
Cross user, no grant	Negative(UNAUTHORIZED)	Negative(UNAUTHORIZED) OR REALM_NOT_TRUSTED	policy denial evidence
System/service caller	system route if role allowed	trusted system delegation	service identity, role proof

12 Stop 状态机器

runtime stop 是 operator lifecycle, 不是 unpair。它必须停止本地 daemon/session 并撤销 namespace 可见性, 但保留 credential、trust 和 pairing identity。



Code	State	Owner	Invariant
0100	STOP_REQUESTED	CLI	user requested runtime stop
0110	STOP_PRECHECK	CLI daemon	load local runtime identity; no network-heavy inference
0200	INVOCATION_DRAINING	CLI daemon	in-flight invocations get completed/cancelled receipts
0210	SESSION_DRAINING	CLI daemon / Axon bridge	stop accepting new remote invocations
0300	NAMESPACE_WITHDRAWING	CLI daemon / Hub	withdraw advertised abilities/presence
0400	SESSION_CLOSING	CLI daemon / Hub	close admitted session with reason operator_stop
0500	DAEMON_STOPPING	CLI supervisor	daemon process exits or is killed after timeout
0900	STOPPED	CLI	daemon stopped; credential remains
F650	STOP_PRECHECK_FAILED	CLI	local runtime identity/config unreadable
F660	DRAIN_TIMEOUT	CLI daemon	pending receipts must be cancelled
F670	SESSION_CLOSE_FAILED	CLI daemon / Hub	local stop may continue but state is degraded
F680	NAMESPACE_WITHDRAW_FAILED	CLI daemon / Hub	namespace visibility may be stale
F690	DAEMON_STOP_FAILED	CLI supervisor	process did not stop

13 Self Uninstall / Leave 状态机器

self uninstall/leave 是 identity lifecycle。它撤销本地 device identity、trust material、cached projections 和 backend pairing, 与 runtime stop 不同。clean target 不兼容旧的静默清理: 部分失败必须可见且可恢复。

Code	State	Owner	Invariant
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U100	UNINSTALL_REQUESTED	CLI	explicit destructive user request
U110	IDENTITY_LOADED	CLI	load device id, realm, credential paths, trust paths
U200	LOCAL_RUNTIME_STOPPING	CLI	reuse stop state machine; no live session may remain advertised
U300	NAMESPACE_REVOKING	CLI daemon / Hub	remove presence/ability advertisements and revoke namespace authority
U400	BACKEND_PAIRING_REVOKING	Backend	mark device pairing revoked/removed with typed reason
U500	LOCAL_CREDENTIALS_REMOVING	CLI	delete device credential/private local config
U600	TRUST_REMOVING	CLI	remove realm trust files only if no other identity needs them
U700	LOCAL_CACHE_PURGING	CLI	purge read model/session/invoke cache for this identity
U900	UNINSTALLED	CLI / Backend	local identity cannot invoke; frontend shows removed/unpaired
F710	IDENTITY_NOT_FOUND	CLI	visible unless user requested allow-missing
F720	UNINSTALL_STOP_FAILED	CLI	do not delete credentials while runtime may still advertise
F730	REVOKE_FAILED_RETRYABLE	CLI / Hub	backend/hub revoke failed; pause or mark partial
F740	CREDENTIAL_REMOVE_FAILED	CLI	local credential deletion failed
F750	TRUST_REMOVE_FAILED	CLI	trust cleanup failed
F760	CACHE_PURGE_FAILED	CLI	cache cleanup failed, identity already revoked
F770	PARTIAL_UNINSTALL	CLI / Backend	irreversible steps completed; doctor shows remaining cleanup

14 Canonical Failure Code Catalog

Pairing/join/runtime

Codes: PAIRING_TOKEN_EXPIRED, PAIRING_TOKEN_CONSUMED, HUB_UNREACHABLE, CREDENTIAL_REJECTED, DAEMON_BOOT_FAILED, BRIDGE_LIBRARY_NOT_FOUND, SESSION_ADMISSION_REJECTED, NAMESPACE_NOT_VISIBLE.

Surfaces: join, doctor, backend device DTO.

Resolver negative

Codes: ABILITY_QUERY_CONFLICT, NXDOMAIN, NODATA, NOROUTE, UNAUTHORIZED, STALE, THROTTLED, OVERLOADED, ROUTE_PROFILE_NOT_AUTHORITATIVE.

Surfaces: resolve answer, read model DTO.

Routing/tenant

Codes: REALM_NOT_TRUSTED, PEER_HUB_UNAVAILABLE, DELEGATION_REQUIRED, DELEGATION_INVALID, SUBJECT_KEY_UNREGISTERED, CALLER_SIGNATURE_INVALID, TARGET_NOT_IN_PRESENCE_REGISTRY, AGENT_NOT_ADVERTISED, AGENT_NOT_RUNNING.

Surfaces: resolver answer, invocation receipt.

Invocation transport

Codes: STREAM_OPEN_FAILED, STREAM_CLOSED, BIDI_OPEN_FAILED, DOWNLOAD_TRAILER_FAILURE, RECEIPT_FAILURE_MISSING, CHILD_INVOCATION_FAILED.

Surfaces: receipt, trailer, WebSocket/control frame.

Stop lifecycle

Codes: STOP_PRECHECK_FAILED, DRAIN_TIMEOUT, SESSION_CLOSE_FAILED, NAMESPACE_WITHDRAW_FAILED, DAEMON_STOP_FAILED.

Surfaces: runtime stop, doctor --json.

Self-uninstall lifecycle

Codes: IDENTITY_NOT_FOUND, UNINSTALL_STOP_FAILED, REVOKE_FAILED_RETRYABLE, CREDENTIAL_REMOVE_FAILED, TRUST_REMOVE_FAILED, CACHE_PURGE_FAILED, PARTIAL_UNINSTALL.

Surfaces: self uninstall/leave, doctor --json.

Decision

Failure code 与状态码不是同一个字段: 状态码定位 state/transition, failure code 解释协议或产品原因。例如 F900 INVOCATION_FAILED 可以携带 TARGET_NOT_IN_PRESENCE_REGISTRY、UNAUTHORIZED 或 AGENT_NOT_ADVERTISED。

15 治本更新计划

1. **Axon contract finalization:** ResolveAnswer 成为唯一 route wire/product contract; Negative 覆盖 deterministic taxonomy; InvocationReceipt.failure 覆盖 Failed/TimedOut/Cancelled。
2. **CLI daemon resolve-before-invoke:** 所有 local/remote invoke path 先 namespace.resolve; dispatch 只消费 resolver-selected next hop; doctor 输出 connection snapshot 和 interrupted transition。
3. **Backend route facade cleanup:** prepare envelope 调 daemon namespace.resolve; 删除长期 canonicalDeviceAbilityRoute 和 owner inference fallback; history/list 返回 typed route failure。
4. **Backend read model typed unavailable:** devices、abilities、agents、skills、pages、calls、history 全部保留 resolve_unavailable[]。
5. **Frontend route source cleanup:** ability action 按 API routes[] 或 selected route id 调用; 下载 client 解码 X-EasyNet-Failure trailer。
6. **Receipt producer consolidation:** terminal/file/browser/remote desktop/openai pages 等 producer 统一写 InvocationReceipt.failure。
7. **End-to-end gates:** docker compose 内覆盖 bridge missing、hub offline、target offline、agent not advertised、unauthorized 五类失败。
8. **Operator lifecycle gates:** runtime stop 使用 0* 状态机; self uninstall/leave 使用 U* 状态机; doctor --json 在 join、invoke、stop、uninstall 中输出同构状态对象: state_code, failure, transition, next_action。

16 剩余任务清单

Priority	Task	Owner repo	Target state
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P0	CLI daemon 所有 invoke path 统一 resolve-before-invoke	EasyNet-Cli	R200 -> R300 -> I400
P0	Backend prepare/submit 删除 canonical ability inference	EasyNet Backend	R300 -> I100
P0	invocations/history/list 改为 resolver catalog route 或 typed failure	Backend/Frontend	no bare 400
P0	meta.list_resources 选择 resolver catalog route	CLI/Backend	no missing route
P1	Frontend 全部 ability action 消费 routes[]	Frontend	no owner inference
P1	下载 client 解码 X-EasyNet-Failure trailer	Frontend	typed download failure
P1	所有 terminal receipt producer 写 InvocationReceipt.failure	CLI/Axon	typed receipt
P1	runtime stop 输出 0* 状态码和 interrupted transition	EasyNet-Cli	stop traceable
P1	self uninstall/leave 输出 U* 状态码和 partial cleanup	CLI/Backend	uninstall traceable
P1	invoke 变体按 route/transport profile 写 fixture	Axon/CLI	complete invoke matrix
P2	RFC-005 failure code taxonomy 细化	Axon	fewer generic failures
P2	Docker e2e 增加 five-failure matrix	all	regression gate

17 验收矩阵

Scenario	Expected state	Expected code
token expired before CLI validate	PAIRING_TOKEN_EXPIRED	F120
join saved credential but Hub TLS unreachable	HUB_UNREACHABLE	F530
bridge library missing	BRIDGE_UNAVAILABLE	F560
daemon up but session admission rejected	SESSION_ADMISSION_REJECTED	F570
device not in PresenceRegistry on terminal.list	ROUTE_NEGATIVE or degraded read model	TARGET_NOT_IN_PRESENCE_REGISTRY
pages agent not advertised	degraded pages read model	AGENT_NOT_ADVERTISED
history/list lacks selected route	typed prepare failure	ABILITY_ROUTE_REQUIRED or NOROUTE
healthy empty pages	render empty with no unavailable	no failure
failed terminal invocation	receipt failure populated	canonical failure code

cross realm untrusted invoke	ROUTE_NEGATIVE	REALM_NOT_TRUSTED
same realm cross-user without grant	ROUTE_NEGATIVE	UNAUTHORIZED
runtime stop success	STOPPED	0900
runtime stop drain timeout	DRAIN_TIMEOUT	F660
self uninstall partial backend revoke failure	PARTIAL_UNINSTALL	F770

18 工程对象边界

Decision

[derived] 干净实现不应通过堆叠 if/else 消灭症状, 而应把状态机、resolver facade、catalog route DTO、failure locator 和 receipt mapper 分成职责小而硬的对象。

ConnectionStateMachine

CLI-owned local join/runtime/namespace snapshot; 决定状态和 transition, 不做 IO。

ResolveRouteFacade

Backend-owned product facade over daemon namespace.resolve; 只返回 typed route/negative/unavailable。

AbilityCatalogRoute

Backend/Frontend shared DTO route item; UI action 的唯一 route source。

FailureLocator

Axon/CLI/Backend shared failure projection; carrier 可随 transport 变化。

InvocationReceiptFailureMapper

runtime producer mapper; transport/runtime/admission errors 到 canonical receipt failure。